Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-16.jpg, we are tasked with evaluating the following limit:

$$L = \lim_{x \to 0} \frac{\sum_{n=1}^{\infty} (-1)^{n-1} \sin(n\pi x)}{\sum_{n=1}^{\infty} (-1)^{n-1} \sin(nex)}$$

1. Evaluating the Infinite Series

First, let us find a closed-form expression for the infinite series present in both the numerator and the denominator. Consider a general series of the form:

$$S(t) = \sum_{n=1}^{\infty} (-1)^{n-1} \sin(nt)$$

We can express this series as the imaginary part of a complex geometric series. Using Euler's formula, $\sin(nt) = \Im(e^{int})$, we have:

$$S(t) = \Im\left(\sum_{n=1}^{\infty} (-1)^{n-1} e^{int}\right)$$

Recognizing the sum as a geometric series with the first term $a = e^{it}$ and common ratio $r = -e^{it}$, we obtain the analytic continuation for the sum:

$$\sum_{n=1}^{\infty} e^{it}(-e^{it})^{n-1} = \frac{e^{it}}{1 - (-e^{it})} = \frac{e^{it}}{1 + e^{it}}$$

To extract the imaginary part, we factor out $e^{it/2}$ from both the numerator and the denominator:

$$\frac{e^{it}}{1 + e^{it}} = \frac{e^{it/2} \cdot e^{it/2}}{e^{it/2}(e^{-it/2} + e^{it/2})} = \frac{e^{it/2}}{2\cos\left(\frac{t}{2}\right)}$$

Applying Euler's formula to $e^{it/2}$ in the numerator:

$$\frac{\cos\left(\frac{t}{2}\right) + i\sin\left(\frac{t}{2}\right)}{2\cos\left(\frac{t}{2}\right)} = \frac{1}{2} + \frac{i}{2}\tan\left(\frac{t}{2}\right)$$

Taking the imaginary part gives us the closed form for our series:

$$S(t) = \Im\left(\frac{1}{2} + \frac{i}{2}\tan\left(\frac{t}{2}\right)\right) = \frac{1}{2}\tan\left(\frac{t}{2}\right)$$

2. Applying the Limit

Now we substitute this closed-form expression back into our original limit. The numerator corresponds to the series evaluated at $t = \pi x$ and the denominator at $t = ex$:

$$L = \lim_{x \to 0} \frac{S(\pi x)}{S(ex)} = \lim_{x \to 0} \frac{\frac{1}{2}\tan\left(\frac{\pi x}{2}\right)}{\frac{1}{2}\tan\left(\frac{ex}{2}\right)}$$

Simplifying the expression by cancelling the $\frac{1}{2}$ factors:

$$L = \lim_{x \to 0} \frac{\tan\left(\frac{\pi x}{2}\right)}{\tan\left(\frac{ex}{2}\right)}$$

Conclusion

To evaluate this final limit as $x \to 0$, we can use the standard small-angle approximation $\tan(\theta) \sim \theta$ as $\theta \to 0$ (or alternatively, apply L'Hôpital's Rule).

$$\tan\left(\frac{\pi x}{2}\right) \approx \frac{\pi x}{2} \quad \text{and} \quad \tan\left(\frac{e x}{2}\right) \approx \frac{e x}{2}$$

Substituting these equivalences into the limit gives:

$$L = \lim_{x \to 0} \frac{\frac{\pi x}{2}}{\frac{e x}{2}} = \frac{\pi}{e}$$

Therefore, the evaluated limit is:

$$\lim_{x \to 0} \frac{\sum_{n=1}^{\infty} (-1)^{n-1} \sin(n\pi x)}{\sum_{n=1}^{\infty} (-1)^{n-1} \sin(nex)} = \frac{\pi}{e}$$